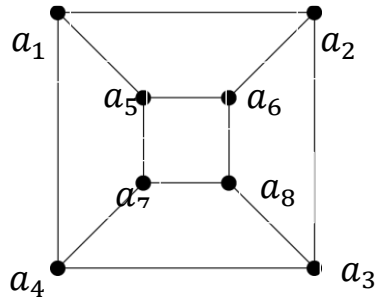
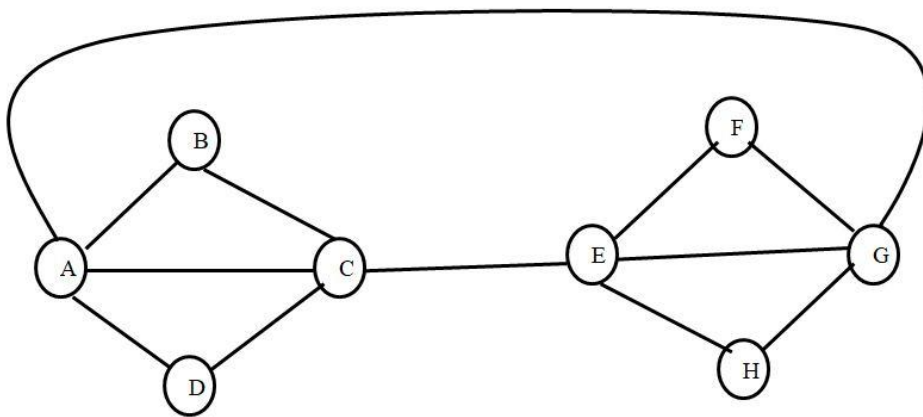


Question Bank Unit-III

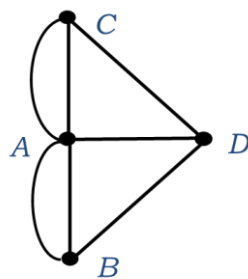
1. Define bipartite Graph. Check whether the following graph is bipartite or not?



2. Define Euler circuit. State the necessary and sufficient condition for a connected graph to have a Euler circuit.
3. Prove that in a simple graph, the number of vertices of odd degree is always even.
4. State and prove Euler's formula for a connected planer simple graph.
5. Use Fleury's algorithm to construct Euler path or circuit in the following graph:

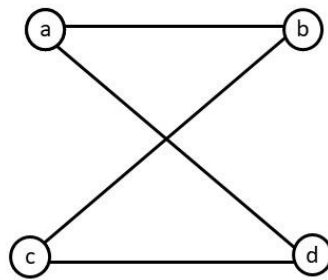


6. Check whether the following graph has a Euler circuit or not.

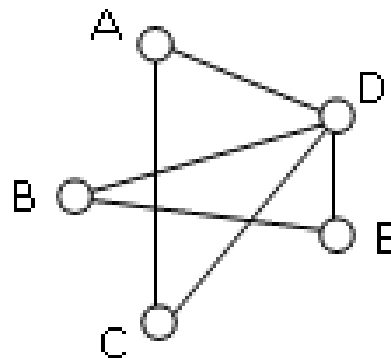


7. Prove that every planar graph has a vertex of degree at most 5.

8. Construct the adjacency matrix of the following graph. How many paths of length four are there from a to d in the graph G .



9. Define Euler circuit. Use Fleury's algorithm to find the Euler Circuit in the following graph



10. Prove that for a connected simple planar graph has a vertex degree that does not exceed 5.

11. Define degree of a vertex in a graph.
12. What is the chromatic number of a complete graph with 4 vertices?
13. What is a Hamiltonian circuit?
14. A graph contains 21 edges, 3 vertices of degree 4 and all other vertices of degree 2. Calculate the total number of vertices in the graph.
15. How many edges does $K_{2,6}$ contain?
16. Find the chromatic number of complete the graph K_5 .
17. What is the chromatic number of an odd cycle.
18. How many edges K_{100} contains?
19. How many edges W_{100} contains?
20. Determine which of the graph is bipartite.
 - a. C_3
 - b. C_5
 - c. C_4
 - d. C_6